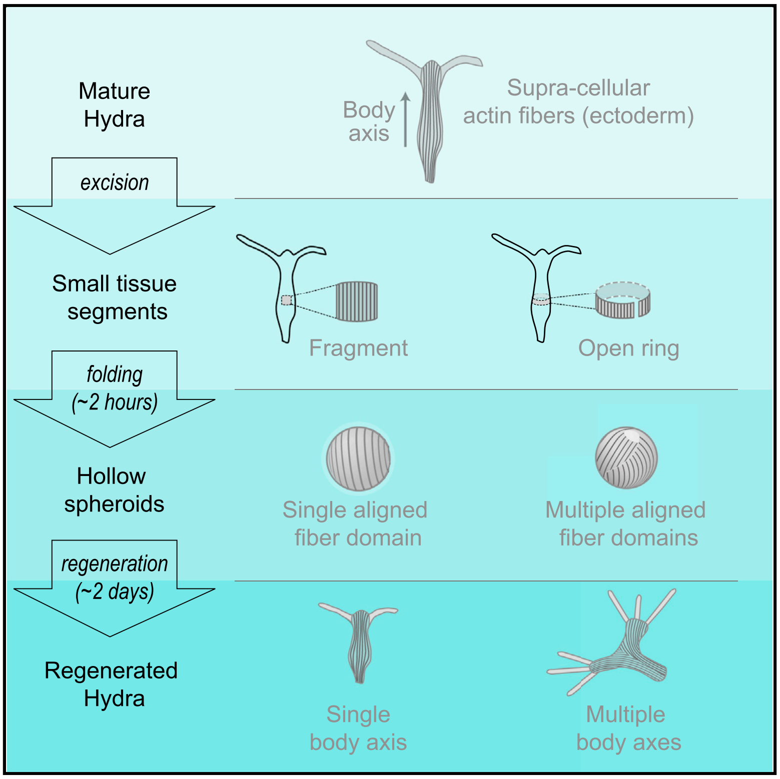
Cell Reports

Article

Structural Inheritance of the Actin Cytoskeletal Organization Determines the Body Axis in Regenerating *Hydra*

Graphical Abstract



Authors

Anton Livshits, Lital Shani-Zerbib, Yonit Maroudas-Sacks, Erez Braun, Kinneret Keren

Correspondence

erez@ph.technion.ac.il (E.B.),
kinneret@physics.technion.ac.il (K.K.)

In Brief

Livshits et al. show that structural inheritance of the supra-cellular actin fiber organization from the parent *Hydra* determines the body axis in regenerating tissue segments. The emergence of multiple body axes is traced to discrepancies in the initial actin fiber alignment and can be suppressed by applying mechanical constraints.

Highlights

- The actin fiber organization in tissue segments is inherited from the parent *Hydra*
- The inherited actin organization determines the body axis in regenerating tissues
- Multiple body axes can be traced to discrepancies in initial actin fiber alignment
- Anchoring regenerating *Hydra* on wires suppresses the emergence of multiple body axes

CrossMark
Livshits et al., 2017, Cell Reports *18*, 1410–1421
February 7, 2017 © 2017 The Author(s).
http://dx.doi.org/10.1016/j.celrep.2017.01.036

CellPress

OPEN ACCESS
CellPress

Cell Reports
Article

Structural Inheritance of the Actin Cytoskeletal Organization Determines the Body Axis in Regenerating *Hydra*

Anton Livshits,[1,4] Lital Shani-Zerbib,[1,4] Yonit Maroudas-Sacks,[1] Erez Braun,[1,2,*] and Kinneret Keren[1,2,3,5,*]
[1]Department of Physics, Technion – Israel Institute of Technology, Haifa 32000, Israel
[2]Network Biology Research Laboratories, Technion – Israel Institute of Technology, Haifa 32000, Israel
[3]Russell Berrie Nanotechnology Institute, Technion – Israel Institute of Technology, Haifa 32000, Israel
[4]Co-first author
[5]Lead Contact
*Correspondence: erez@ph.technion.ac.il (E.B.), kinneret@physics.technion.ac.il (K.K.)
http://dx.doi.org/10.1016/j.celrep.2017.01.036

SUMMARY

Understanding how mechanics complement bio-signaling in defining patterns during morphogenesis is an outstanding challenge. Here, we utilize the multicellular polyp *Hydra* to investigate the role of the actomyosin cytoskeleton in morphogenesis. We find that the supra-cellular actin fiber organization is inherited from the parent *Hydra* and determines the body axis in regenerating tissue segments. This form of structural inheritance is non-trivial because of the tissue folding and dynamic actin reorganization involved. We further show that the emergence of multiple body axes can be traced to discrepancies in actin fiber alignment at early stages of the regeneration process. Mechanical constraints induced by anchoring regenerating *Hydra* on stiff wires suppressed the emergence of multiple body axes, highlighting the importance of mechanical feedbacks in defining and stabilizing the body axis. Together, these results constitute an important step toward the development of an integrated view of morphogenesis that incorporates mechanics.

INTRODUCTION

Morphogenesis, the emergence of well-defined patterns of functional tissues during development, is carried out by the collective dynamics of multiple processes that encompass different levels of organization, from local molecular events to large-scale morphogenetic fields (Forgacs and Newman, 2005; Gilbert, 2000). This patterning involves both the determination of the overall body plan and the detailed specification of tissue types and organ formation via cell differentiation (Gilbert, 2000). The main challenge in morphogenesis research is understanding how all the different processes are integrated to form robust, large-scale, spatio-temporal patterns under diverse and varying environmental and initial conditions. A growing number of examples show that physical processes, in particular, mechanical effects, are an integral part of this patterning process (Beloussov, 2008; Brouzés and Farge, 2004; Davidson, 2012; Goehring and Grill, 2013; Guillot and Lecuit, 2013; Heisenberg and Bellaïche, 2013; Howard et al., 2011; Thompson, 1942; Urdy, 2012). For example, mechanical forces generated by the actomyosin cytoskeleton (Munjal and Lecuit, 2014; Munjal et al., 2015) or changes in cell volume (Saias et al., 2015) are essential for shaping developing embryos. The resulting tissue deformations can also provide mechanical cues that activate biochemical signaling pathways (Pouille et al., 2009), induce gene expression (Desprat et al., 2008), or direct cell fate (Miller and Davidson, 2013). Despite this progress, an integrated view of the role of mechanical processes throughout morphogenesis is still fundamentally lacking. Here, we take advantage of *Hydra*, a small predatory fresh water animal famous for its extraordinary regeneration capabilities, to demonstrate that cytoskeletal organization and mechanical feedbacks play a central role in defining and stabilizing the body axis.

Hydra is a multicellular organism of the Cnidarian phylum, which includes jellyfish and corals. It has $\sim 10^5$ cells belonging to approximately ten different cell types (Bode et al., 1973; Galliot, 2013). Typically, *Hydra* reproduces asexually by producing a bud, but they can also reproduce sexually under stressful conditions (Galliot, 2013). *Hydra* has remarkable regeneration abilities. An adult *Hydra* can regenerate from a small tissue segment (>2% of the adult animal) (Bode and Bode, 1980; Bode and Bode, 1984) or even from a condensed aggregate formed from a mixture of completely dissociated cells (Gierer et al., 1972). *Hydra*'s rapid and flexible morphogenesis, together with its simple body plan and ease of experimental manipulations, make it an excellent model system for studying fundamental aspects of animal morphogenesis (Galliot, 2012; Lenhoff et al., 1986). In fact, historically, many of the important conceptual advances in morphogenesis research, such as the role of an "organizer," the idea of positional information, and the idea of pattern formation by reaction-diffusion dynamics of morphogens, were initially inspired by experimental investigations on *Hydra* (Bode, 2012; Browne, 1909; Gierer and Meinhardt, 1972; Meinhardt, 1993; Turing, 1952; Wolpert et al., 1972).

The body of an adult *Hydra* is shaped as a hollow tube, ~1 cm in length, with a head on one end (Bode, 2009). The head has an opening that forms the *Hydra*'s mouth (called hypostome) and is surrounded by tentacles responsible for catching prey. The opposite end of the tube forms the foot (peduncle), which can attach to surfaces. The *Hydra*'s body is made of a bilayer of epithelial sheets, with a monolayer of endoderm cells on the inner side of the tube surrounded by a monolayer of ectoderm cells. The two epithelial layers are connected to an elastic extracellular matrix, called the mesoglea, situated between the two layers (Sarras, 2012). Within each layer, the actomyosin cytoskeleton forms cortical networks, which are characteristic of epithelial sheets. The cytoskeleton also forms supra-cellular networks of contractile fibers, which span the whole organism, near the basal side of each layer (Kass-Simon, 1976; Mueller, 1950; Seybold et al., 2016; West, 1978). These actomyosin fibers are organized along the animal axis in the ectoderm and in a perpendicular (circumferential) direction in the endoderm. The fibers are connected between neighboring cells within the epithelial layers via cell-cell junctions and are mechanically coupled to the mesoglea matrix (Seybold et al., 2016). This cytoskeletal organization is essential for the physiology of a mature *Hydra* because the contraction of the ectoderm or endoderm actomyosin fibers leads, respectively, to shrinkage or expansion of the animal body. However, the role of the actomyosin cytoskeleton during *Hydra* morphogenesis is still largely unexplored (Seybold et al., 2016).

Hydra regeneration provides a unique opportunity to study morphogenesis from a variety of initial conditions (Bode and Bode, 1980; Gierer et al., 1972). Importantly, it is possible to follow the whole morphogenesis process in individual regenerating *Hydra* to get a holistic view of the pattern formation process (Gierer, 2012; Gierer et al., 1972; Hobmayer et al., 2000; Seybold et al., 2016). The initial step in regeneration from tissue segments is the folding of the bilayer into a hollow spheroid (Bode and Bode, 1984; Krahe et al., 2013). The resulting spheroids undergo extensive shape fluctuations as they develop into small *Hydra*. This regenerative process mostly involves cell rearrangements and differentiation, whereas cell division does not seem to be essential (Cummings and Bode, 1984; Gierer and Meinhardt, 1972; Park et al., 1970). The morphological dynamics during regeneration from tissue fragments can be divided into two phases: an initial phase with large-amplitude, osmotically driven expansion-contraction cycles, followed by a second phase characterized by rapid, more erratic, smaller amplitude fluctuations (Fütterer et al., 2003; Kücken et al., 2008; Wanek et al., 1980). The transition between these phases was interpreted to be the signature of a symmetry-breaking event, in which the new body axis of the regenerating *Hydra* is defined (Fütterer et al., 2003; Soriano et al., 2009). However, the mechanism involved in specifying the body axis in regenerating tissue segments remains unclear.

In this work, we study *Hydra* regeneration from tissue segments and show that the actomyosin cytoskeleton is a key ingredient in the regenerative process. We find that the alignment of the supra-cellular contractile actomyosin fibers in excised tissue segments confers a memory that persists throughout the regeneration process and specifies the body axis in the newly formed *Hydra*. These results also show that axis determination in regenerating *Hydra* tissue fragments does not involve spontaneous symmetry breaking, as previously assumed (Fütterer et al., 2003; Soriano et al., 2009), but rather is inherited from the parent *Hydra* through the organization of the actin cytoskeleton. Moreover, by varying the initial geometry of the excised tissue, we find that tissue spheroids containing multiple regions with distinct actin fiber alignment at the onset of regeneration form animals with multiple body axes (i.e., more than one head or foot). We further show that anchoring regenerating tissues on stiff wires promotes order in the regeneration process because it increases the odds of forming normal *Hydra* with a single body axis. Together, our results illustrate the importance of endogenous actomyosin force generation and mechanical feedbacks during axis formation and stabilization in regenerating *Hydra*.

RESULTS

Regeneration Axis Is Inherited via the Organization of the Actin Cytoskeleton

Excised tissue fragments from the gastric region of mature *Hydra* regenerate into viable *Hydra* within a couple of days (Figure S1; Movie S1). During the first 1–3 hr after excision, the tissue folds into a closed hollow spheroid, with the ectoderm layer on the outside. Subsequently, the spheroid undergoes dynamic shape changes and eventually assumes the morphology of an adult *Hydra* (Figure S1; Movie S1). Regeneration occurs in fragments above a minimal size (Shimizu et al., 1993). Smaller tissue segments are able to close into spheroids but disintegrate within hours, whereas the vast majority of larger spheroids regenerate into viable animals (~88%; n = 102, for radius >100 μm). The most unequivocal morphological feature of regeneration is the appearance of tentacles, followed by the development of a cone-shaped hipostome between them. We define the regeneration time as the time between tissue excision and the first appearance of a tentacle. The average regeneration time for tissue fragments is 35 ± 9 hr (mean ± SD; n = 89), and is only weakly dependent on size (Figure S2A).

The initial organization of the actomyosin cytoskeleton in an excised tissue fragment is related to the cytoskeletal organization in the parent *Hydra* prior to excision. The most prominent cytoskeletal feature in mature *Hydra* is the actomyosin fibers near the basal side of the ectoderm layer, which are aligned parallel to the animal axis (Figure 1A) (Kass-Simon, 1976; Mueller, 1950; Seybold et al., 2016; West, 1978). Analysis of the actin cytoskeleton in tissue fragments revealed that this actin organization is preserved following excision (Figure 1B). Thus, excised tissue fragments contain an array of parallel ectodermal actin fibers, which are inherited from the parent *Hydra* and aligned with its body axis.

The actin dynamics throughout the regeneration process can be followed in live animals using transgenic *Hydra* expressing GFP-lifeact in the ectoderm, which efficiently labels the ectodermal actin fibers (Seybold et al., 2016) (Figure 1D). During the folding process, the initially square-shaped fragment excised from the side of the tubular *Hydra* body folds into a closed, hollow spheroid (Figures 1C and 1D). The actin fibers persist during

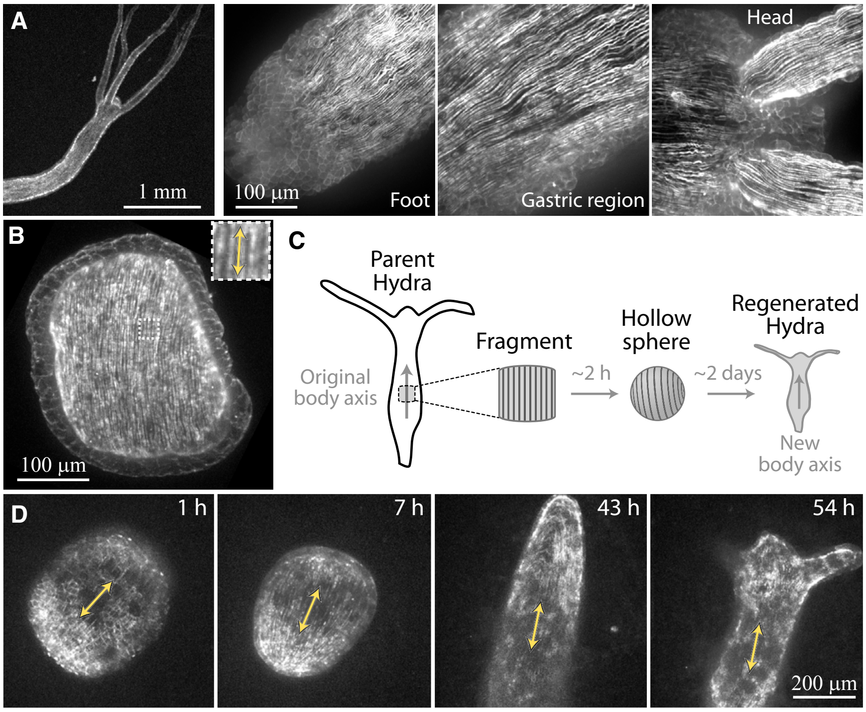


Figure 1. Dynamic Organization of the Actin Cytoskeleton during Regeneration from Tissue Fragments

(A) Fluorescence images depicting the supracellular organization of the actin cytoskeleton in mature *Hydra* expressing lifeact-GFP in the ectoderm. Maximum intensity projections of spinning-disk confocal z stacks show the whole *Hydra* (left) and zoomed views of regions in the foot, gastric region, and head (right). The most prominent cytoskeletal features are the longitudinal actomyosin fibers, which span the entire animal, from head to foot, along the body axis.

(B) Maximum projection image of a spinning-disk confocal z stack of a fragment, which was fixed and stained with AlexaFluor 546 phalloidin immediately after excision. Inset: a zoomed view of the aligned ectodermal actin fibers, which were oriented parallel to the original body axis of the parent *Hydra* prior to excision.

(C) Schematic illustration of the regeneration experiment. A tissue fragment from the side of the gastric region of a mature *Hydra* is excised. The fragment folds into a spheroid within a couple of hours and regenerates into a full *Hydra* within a couple of days.

(D) The regeneration of a tissue fragment excised from a transgenic *Hydra* expressing GFP-lifeact in the ectoderm is followed by spinning disk confocal microscopy (Movie S2). Maximum projection images from confocal z stacks at different time points are shown. The arrows indicate the alignment of actin fibers. See also Figure S1 and Movies S1 and S2.

the folding process so the resulting hollow spheroid also contains aligned fibers. Subsequently, the regenerating spheroid exhibits substantial shape changes, yet the initial axis, determined by the orientation of the ectodermal actin fibers at the onset of regeneration, is preserved throughout the entire process (Figure 1D; Movie S2). Remarkably, the orientation of the ectodermal actomyosin fibers defines the new animal axis, as illustrated by the appearance of tentacles along the fibers' direction in the regenerating *Hydra*. Thus, we find that the actin organization in excised tissue fragments is inherited from the parent *Hydra* and aligned with the body axis of the regenerated animal. This also implies that the concept of symmetry breaking previously used to describe *Hydra* regeneration from tissue fragments (Fütterer et al., 2003; Soriano et al., 2009) is inaccurate because excised fragments obviously have a pronounced structural asymmetry carried over from the parent organism.

The importance of active force generation by the actomyosin cytoskeleton during *Hydra* regeneration was examined by pharmacological perturbations of the contractile activity of myosin motors (Figure S3). Blocking myosin activity directly with blebbistatin (Straight et al., 2003) at high doses (25 μM) inhibits tissue folding and prevents the formation of closed spheroids. Treated fragments remain flat and disintegrate within hours (Figure S3B). Because the formation of a closed spheroid seems essential for the physiology of regenerating *Hydra*, we tested the influence of myosin perturbations applied 3 hr after excision, allowing excised tissue to undergo the initial folding and form a closed spheroid prior to the perturbation. Both blebbistatin, which inhibits myosin II activity directly, and ML7, which interferes with myosin activity by inhibiting myosin light chain kinase (Saitoh et al., 1987), completely prevent regeneration of folded spheroids (Figures S3A and S3C). Interestingly, regeneration is even blocked by much lower doses of blebbistatin (1 μM), which confer only partial inhibition of myosin activity. Under these conditions, myosin can still drive the initial tissue folding into closed spheroids (Figure S3B), and treated tissues remain viable for a longer time (37 ± 8 hr, mean ± SD; n = 16), yet regeneration is still completely blocked (Figures S3A and S3C). Blebbistatin-treated tissue retains a perfectly spherical shape and fails to elongate or develop tentacles (Figure S3C). The Rho kinase inhibitor Y-27632 did not have a noticeable effect on *Hydra* regeneration (at concentrations up to 25 μM; Figures S3A and S3C), suggesting that the Rho pathway is not important during regeneration or that this compound is ineffective in *Hydra*. We also tried to enhance myosin activity by treating a regenerating *Hydra* with calyculin A, a phosphatase inhibitor that promotes myosin contraction (Ishihara et al., 1989). At doses that did not compromise tissue viability, the regeneration process appeared normal (Figures S3A and S3C).

Together, these pharmacological perturbations suggest that actomyosin force generation is not merely required to drive the initial tissue folding, but also has functional significance during subsequent steps in the regeneration process. However, given the central role of the actomyosin cytoskeleton in many essential

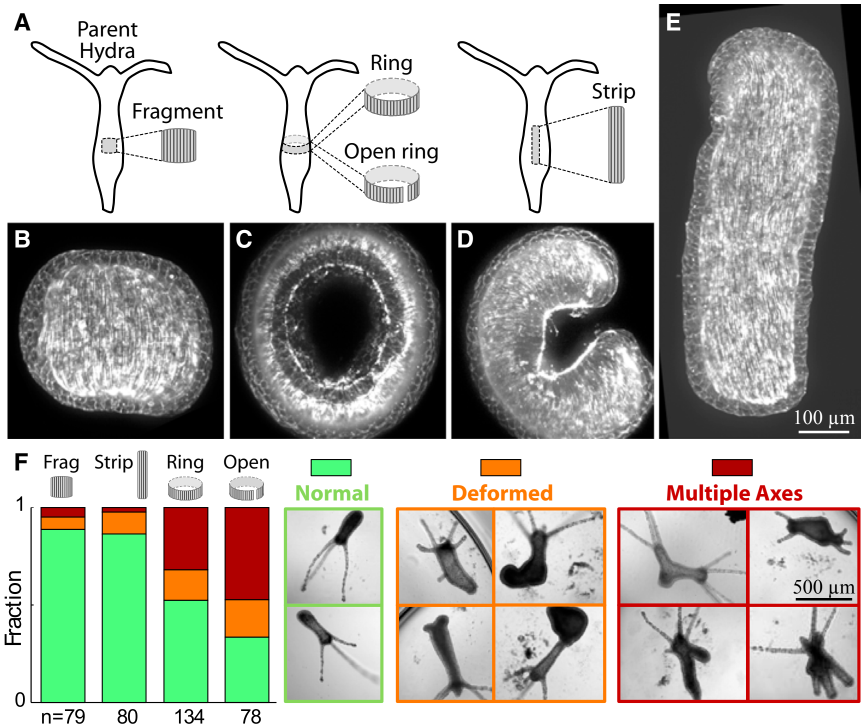


Figure 2. Dependence of the Regeneration Process on Initial Tissue Geometry

(A) Schematic illustration of the different initial tissue segment geometries used in the regeneration experiments.

(B–E) Maximum projection images of spinning-disk confocal z stacks of a fragment (B), a ring (C), an open ring (D), and a longitudinal strip (E). All samples were fixed and stained with phalloidin immediately after excision.

(F) The regeneration outcome as a function of initial tissue shape. Left: bar plot showing the fraction of regenerating *Hydra* in each category of morphological outcome as a function of the initial tissue geometry. Rings and open rings are statistically more likely to regenerate into animals that are deformed or have multiple body axes compared to fragments or longitudinal strips (Chi-squared homogeneity test, p value $< 10^{-6}$). Right: bright-field images showing examples of the different categories of morphological outcome: (1) *Hydra* with normal morphology, (2) deformed *Hydra*, and (3) *Hydra* with multiple body axes. See also Figure S2.

physiological processes, it is difficult to isolate the effect of actomyosin activity on morphogenesis and rule out the possibility that regeneration is prevented because of indirect detrimental effects of myosin inhibition.

Regeneration Depends on Initial Tissue Geometry

We sought to examine in more detail how initial tissue geometry and cytoskeletal organization influence morphogenesis. To this end, we took advantage of the unique flexibility in defining initial conditions in *Hydra* to characterize the regeneration process and its morphological outcome as a function of the shape of the excised tissue segments (Figure 2). Specifically, we consider four different tissue segment geometries, all excised from the gastric region of mature *Hydra* (Figure 2A): fragments (square segments; Figure 2B), rings (thin slices of the tubular body; Figure 2C), open rings (rings that are further cut open; Figure 2D), and longitudinal strips (strips that are elongated along the body axis; Figure 2E). In all the cases, the tissue segments first fold into closed hollow spheroids and subsequently regenerate with similar regeneration times (Figure S2). However, the regeneration dynamics and morphological outcome are dramatically dependent on the shape of the initial tissue segment (Figure 2F). The vast majority of fragments and longitudinal strips regenerate into *Hydra* that display a normal morphology characterized by a single body axis, with a head on one end and a foot on the other. However, the probability to regenerate animals with multiple axes (i.e., more than one head and/or foot) is significantly greater in rings and even more dramatic in open rings. In particular, the majority of open rings regenerate into abnormal morphologies, with nearly half of the regenerated animals exhibiting multiple axes (Figure 2F).

The first step in regeneration is the rapid (1–3 hr) tissue folding, which morphs the excised tissue piece into a closed hollow spheroid. The formation of the spheroid occurs mostly by bending and deforming the epithelial bilayer (so that the ectoderm remains on the outer side) and the adherence of cells at the open edges to form a continuous sheet. This morphing necessarily implies that the tissue layout in the folded spheroid is different from the original organization of the excised tissue in the parent animal. Our observation that the regeneration outcome is strongly dependent on initial tissue geometry (Figure 2F) motivated us to look more closely at the folding process and its influence on actin organization and morphology for different initial conditions.

Excised fragments initially consist of a nearly flat square piece of epithelial tissue (Figure 2B). The tissue bends into a convex surface, with the ectoderm on its outer side, which closes into a spheroid in a "purse-string"-like manner. Thus, the periphery of the fragment comes together, driving tissue parts that were not adjacent in the original animal next to each other in the closed spheroid (Figure 3). These changes in overall tissue morphology also entail a dramatic reorganization of the actin cytoskeleton (Figure 3A; Movie S3). We find that the actin fibers at the center of the fragment maintain their alignment and persist as the tissue folds into a spheroid, whereas the actin cytoskeleton near the periphery of the original tissue undergoes extensive rearrangements and the original fiber organization is lost (Figures 3B and 3C; Movie S4). Over time, the actin cytoskeleton reorganizes in the closure region, and actin fibers that are aligned with the new body axis form (Figure 1D; Movie S2). Thus, the actin fibers exhibit both stability and flexibility. The fibers at the center of the excised fragment retain the memory of their alignment and specify the body axis of the regenerating animal, whereas the actin near the closure region adapts by undergoing extensive rearrangements.

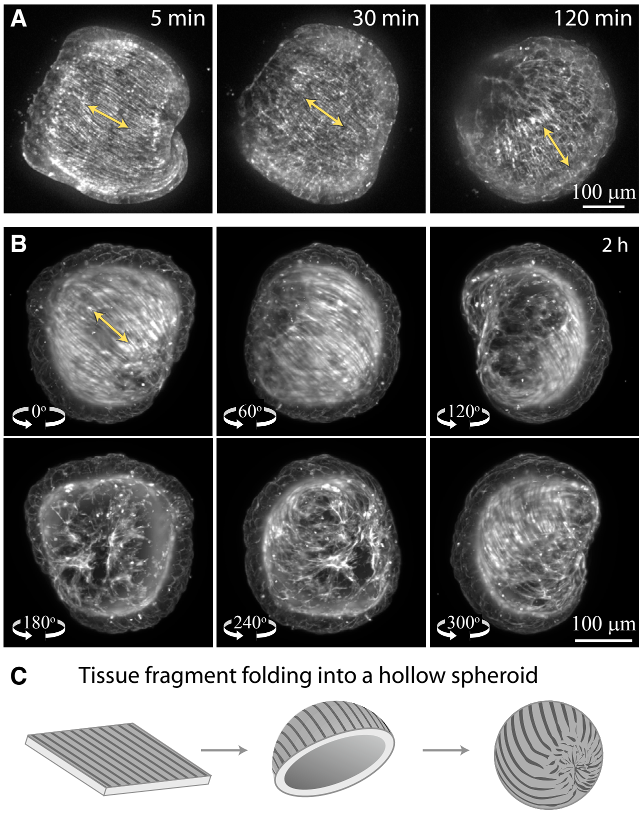


Figure 3. Actin Reorganization during the Folding of an Excised Tissue Fragment into a Spheroid

(A) The actin dynamics during the initial folding of a tissue fragment excised from a transgenic *Hydra* expressing GFP-lifeact in the ectoderm are followed by spinning disk confocal microscopy (Movie S3). Maximum projection images from confocal z stacks at different time points are shown. The arrows indicate the alignment of actin fibers.

(B) Maximum projection images of light sheet z stacks of a tissue fragment folded into a spheroid, ~2 hr after excision. The tissue was fixed and stained with AlexaFluor 546 phalloidin and imaged by light sheet microscopy from six different angles. The maximum projection images from each angle show different facets of the tissue spheroid. The entire z stack from front to rear (0° and 180°) of this spheroid is available (Movie S4). Aligned ectodermal fibers are visible at the front (0°; the region at the center of the original tissue), but not near the closure region at the rear (180°; the region where the periphery of the fragment comes together to form a closed spheroid).

(C) Schematic illustration of the reorganization of the ectodermal actomyosin fibers during the folding of a tissue fragment into a hollow spheroid. The nearly flat square tissue fragment excised from the side of the gastric region of a mature *Hydra* starts folding into a cap, with the ectoderm layer on the outside. The tissue closes into a spheroid in a "purse-string"-like manner, with the actin fibers organized in the inner region and disorganized near the closure region. See also Movies S3 and S4.

Longitudinal strips start off as nearly flat elongated tissue segments, with ectodermal actin fibers running parallel to their long axis (Figure 2E). Within minutes, the strips fold along their long axis, bringing the head side and the foot side together (Figures 4A and 4B; Movie S5). Subsequently, as the strip morphs into a hollow spheroid, cells that were closest to the head in the parent animal adhere to cells that were closest to the foot (Figures 4A and 4B). Thus, even though the excised strip has clear polarity with respect to the animal axis of the parent animal, the resulting spheroid loses this head-foot polarity, and, in particular, does not have a "head" side and a "foot" side. As in fragments, the actin fibers in the central part of the strip remain intact during the folding process and form a single domain of parallel fibers in the folded spheroid (which is often elongated). Time-lapse imaging shows that the new animal axis in the regenerating *Hydra* will form along the direction of the inherited actin fibers (Figure 4C; Movie S6). Thus, in both fragments and strips, the folded spheroid contains a single dominant domain of parallel actin fibers inherited from the parent animal, which persists throughout the regeneration process. Subsequently, in both cases, the majority of excised tissues regenerate into *Hydra* having a normal morphology, with a single body axis that is aligned with the inherited actin fibers.

To pursue the underlying cause for the markedly higher fraction of multiple axes and deformations in animals regenerating from open rings, we follow their actin organization during regeneration (Figure 5; Movie S7). We find a striking correlation between the development of multiple body axes and the initial actin fiber organization (Figures 5 and S4). Early in regeneration, the actin fibers in spheroids that originate from open rings lack global alignment compared to spheroids generated from fragments or longitudinal strips. In particular, open rings often fold asymmetrically, with the two sides of the sliced ring moving in opposite directions, resulting in regions with mismatched fiber orientations (Figure 5; Movie S7 and Movie S8). Thus, the initial spheroid generated from an open ring typically contains regions with distinct ectodermal fiber orientations (Figure 5D). Time-lapse imaging reveals that spheroids that contain multiple regions with distinct actin fiber alignment at the onset of the regeneration process, develop bifurcations in their actin fiber organization and regenerate into animals with multiple body axes (Figures 5C and S4D; Movie S7).

Regenerating closed rings exhibit more abnormalities than do fragments and strips, yet significantly fewer abnormalities than do open rings (Figure 2F). This is reflected in the folding of rings, which start as hollow cylinders and fold into closed spheroids mainly in one of two ways (Figures S4A and S4B). In the first scenario, the top and bottom faces of the open cylinder close, forming a rosette-like structure (Figure S4A). In this case, the folded spheroid essentially maintains the polarity and fiber organization that it had in the parent animal and is thus likely to regenerate into an animal with normal morphology. In the second scenario, the ring buckles, as previously described (Krahe et al., 2013), and closes into a spheroid, in which the tissue is substantially deformed compared to its original layout in the parent animal (Figure S4B). In this case, the resulting spheroid contains regions with distinct ectodermal fiber orientations as in spheroids formed from open rings and will likely regenerate abnormally.

Mechanical Constraints Enhance Order in the Regeneration Process

To examine the role of mechanical feedbacks on morphogenesis, we developed a methodology to anchor regenerating *Hydra* tissue on thin stiff wires (Figure 6; see Supplemental Experimental Procedures). The wire sets a fixed reference axis in the

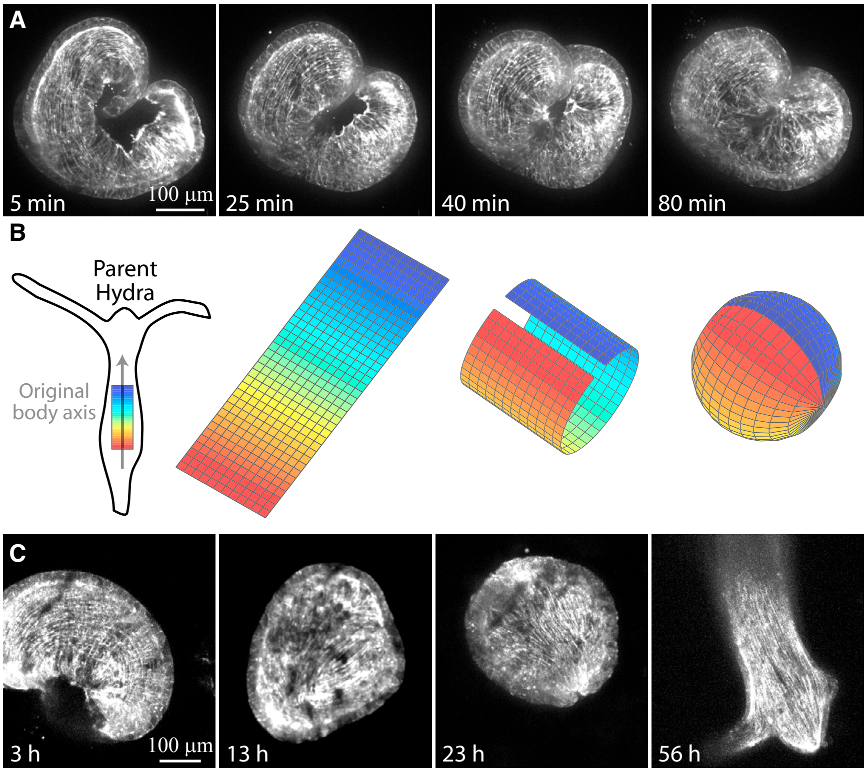


Figure 4. Folding and Regeneration of Longitudinal Strips

(A) The folding of a longitudinal strip excised from a transgenic *Hydra* expressing GFP-lifeact in the ectoderm is followed by time-lapse spinning disk confocal microscopy (Movie S5). Maximum projection images from z stacks at different time points are shown.

(B) Schematic illustration of the folding of a longitudinal strip into a hollow spheroid. The tissue strip is color coded as a function of position in the parent animal (blue is closest to the head, and red is closest to the foot). The excised strip initially folds along its long axis, bringing together cells that were closest to the head (blue) with cells that were closest to the foot (red). Subsequently, the open tissue edges adhere to each other to form a closed spheroid. Note that the resulting spheroid does not have a clear polarity with respect to the original body axis.

(C) The regeneration of a longitudinal strip excised from a transgenic *Hydra* expressing GFP-lifeact in the ectoderm is followed by time-lapse spinning disk confocal microscopy (Movie S6). Maximum projection images from z stacks of the regenerating strip at different time points are shown. The strip regenerates into an animal with a single body axis that is aligned with the initial actin fiber orientation on the spheroid. See also Movies S5 and S6.

system, so the body axis of the regenerated animal can be related to the original tissue orientation. More importantly, the stiff wires induce mechanical constraints on the regenerating tissue. To generate anchored rings, we insert a 25-μm-diameter platinum wire through the gastric region of a mature *Hydra*, perpendicular to its body axis, and cut a thin slice of tissue around the wire (Figure 6A). An anchored open ring can be generated by cutting the ring open at one position (Figure 6A). Alternatively, we can anchor a tissue ring on a wire parallel to the original body axis by threading an excised ring on a wire (Figure 6A). In all cases, the anchored tissue folds into a hollow spheroid on the wire, and the regeneration process proceeds with the wire pierced through the regenerating tissue (Figure 6B; Movie S9). More than 90% of the anchored tissue segments remain viable and regenerate on the wires (n = 165; Figure S2). We find that the regeneration axis is perpendicular to the wire in the perpendicular configuration (Figure 6B) and along the wire in the parallel configuration (Figure 6C). Similarly, the initial actin fiber orientation in the spheroid is perpendicular or parallel to the wire in the perpendicular or parallel configuration, respectively, reflecting the original actin fiber orientation in the parent animal (Figures 6D and 6E). Thus, the wires provide a means to anchor regenerating tissue in a particular orientation relative to the original animal axis and apply mechanical constraints on the tissue without disrupting the regeneration process or directly specifying the regeneration axis.

Surprisingly, we find that the presence of a wire enhances order during the regeneration process (Figure 7A). The fraction of multi-axes and deformed animals is significantly reduced from 48% in non-anchored rings (n = 134) to 16% in rings, which are anchored on wires in the perpendicular configuration (n = 80; p value $< 10^{-4}$, Chi-squared homogeneity test) and 22% in rings in the parallel configuration (n = 74; p value $< 10^{-3}$). Similarly, the fraction of open rings that regenerate into an animal with abnormal morphology goes down from 67% in non-anchored tissues (n = 78) to 34% with wires (n = 70; p value $< 10^{-3}$). The presence of a wire is also found to decrease the average regeneration times (Figure S2), providing an additional indication for the organizing effect induced by the wire. Thus, starting from initial tissue segments with a similar size, shape, and composition, the presence of an anchoring wire promotes proper regeneration into a *Hydra* with a single body axis, which has a head at one end and a foot at the opposite end.

How does anchoring regenerating *Hydra* on wires promote the development of a normal morphology with a single body axis? To address this question, we tracked by live imaging the regeneration of anchored and non-anchored open rings because they had the highest rates of abnormal regeneration (Figures 7C and S1; Movie S9). The changes in morphology during regeneration are characterized by extracting orthogonal modes of shape variability from the contours of the 2D-projected shapes using principal components analysis (see Supplemental Experimental Procedures). These modes concisely capture most of the shape variation over time using a small number of parameters (Pincus and Theriot, 2007). We find that the extracted shape modes are more ordered in the presence of the wire (Figure 7C). In particular, the primary shape mode describing an anchored open ring, which captures ~80% of its shape variation, exhibits

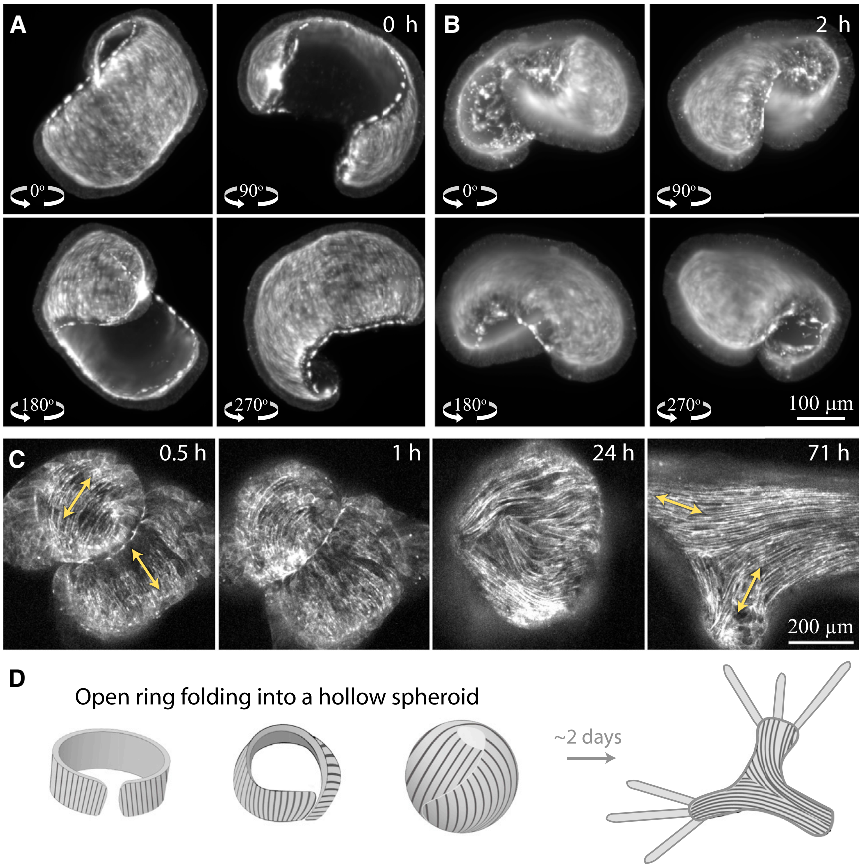


Figure 5. Dynamic Organization of the Actin Cytoskeleton during Regeneration from Open Rings

(A and B) Maximum projection images of light sheet z stack of an open ring taken from four different angles. The tissue was fixed and stained with AlexaFluor 647 phalloidin a few minutes after excision (A) or ~2 hr after excision (B).

(C) The regeneration of an open ring excised from a transgenic *Hydra* expressing GFP-lifeact in the ectoderm is followed by time-lapse spinning disk confocal microscopy (Movie S7). Maximum projection images from z stacks of a regenerating open ring at different time points are shown. The arrows indicate the alignment of actin fibers.

(D) Schematic illustration of the folding of an open ring into a hollow spheroid. Open rings typically fold by moving the two free flaps of the ring in opposite directions, which leads to distinct actin fiber organization at different regions of the folded spheroid (left). Subsequently, such spheroids often regenerate into deformed animals as the two-headed *Hydra* depicted (right). See also Figure S4 and Movies S7 and S8.

deformations primarily along a single axis. This stands in contrast to the shape modes of non-anchored regenerating open rings, which are considerably more irregular and usually do not display a preferred axis. The influence of the wire on the observed shape modes can be rationalized by considering the mechanical constraints induced by the wire (Figure 7B). Shape deformations for which the tissue at the intersection point moves in-plane (i.e., perpendicular to the wire) imply that the wire has to move within the epithelial bilayer. The integrity of the bilayer is expected to hinder such movements and thus favors sliding along the wire. As a result, tissue deformations, in which the two intersection points slide along the wire, will be favored, whereas other shape changes will be suppressed. This bias is reflected in the observed shape modes that vary primarily along a single axis (Figure 7C).

We hypothesize that the dynamics of tissue-shape fluctuations might feed back and influence actin fiber organization. To test this idea, we examined actin fiber dynamics in open rings anchored on wires (Figures 7D–7F; Movie S10) and compared them to non-anchored open rings (Figure 5). The initial folding of an open ring into a spheroid over the first couple of hours of the regeneration process was not strongly influenced by the presence of the wire. Most importantly, the two flaps of anchored open rings often move in opposite directions (Figure 7D), similar to the folding of non-anchored open rings. Thus, the anchored hollow spheroid formed typically contains distinct regions, in which the fibers are oriented in different directions (Figure 7E), as frequently observed in non-anchored open rings. However, over time, the fibers become better aligned with respect to each other and typically become perpendicular to the wire (Figure 7F; Movie S10). Thus, we observe that the presence of the wire promotes the subsequent reorganization of the actin fibers in a dynamic manner. These results suggest that the cytoskeletal organization is not merely a marker of the newly formed body axis in the regenerating animal, but rather plays an active role in defining and stabilizing the body axis via its dynamics throughout morphogenesis.

DISCUSSION

By following the actin organization in regenerating *Hydra*, we expose the importance of cytoskeletal self-organization and mechanical feedbacks in determining the body plan. Our main result is that the regeneration axis is structurally inherited via the large-scale organization of the cytoskeleton. This type of structural inheritance at the supra-cellular level is highly non-trivial because of the dramatic shape changes and tissue folding that occur during morphogenesis. The excised tissue segments carry a memory of the original body axis, which is inherited from the parent animal. However, this well-defined axis inevitably morphs as the tissue folds on itself and closes to form a hollow spheroid. Specifically, although a tissue fragment or strip initially has a parallel array of actin fibers (aligned with the original body axis), after folding the resulting spheroid only has partial fiber alignment because of the closure process. This incomplete organization is sufficient to stimulate order and reorganize the actin fibers dynamically throughout the regenerating tissue. The induced alignment of actin fibers organizes and stabilizes the body axis and directs morphogenesis.

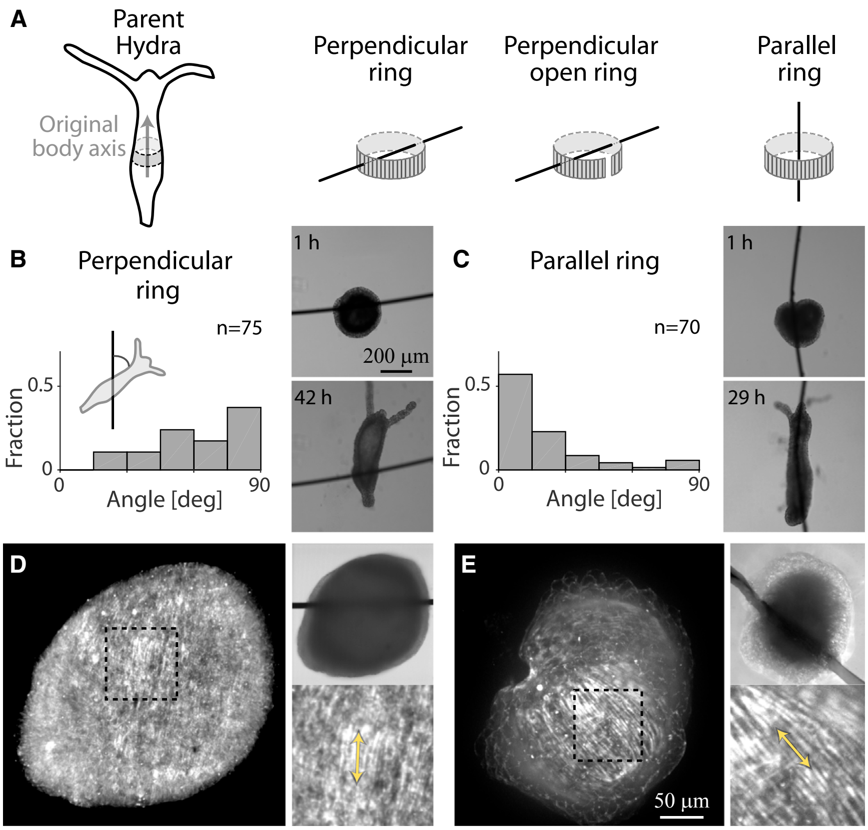


Figure 6. Regeneration of *Hydra* Tissues Anchored on Wires

(A) Schematic illustration of the anchoring of *Hydra* tissues on stiff wires. Perpendicular rings are generated by skewering the parent *Hydra* on a wire, perpendicular to its body axis, and cutting a ring around the wire. An open ring is generated by cutting the excised ring open. Parallel anchored rings are prepared by threading a freshly excised ring on a wire parallel to the original axis.

(B and C) Right: bright-field images of a regenerating hydra ring anchored on a wire perpendicular (B) or parallel (C) to the original animal axis. Images are shown early (top) and late (bottom) in the regeneration process. Left: histograms of the angle of the regeneration axis relative to the wire for rings anchored perpendicular (B) or parallel (C) to the original animal axis. Parallel rings regenerate along the wire, whereas perpendicular rings regenerate perpendicular to the wire.

(D and E) Maximum projection images from spinning disk confocal z stacks of regenerating rings anchored in a perpendicular (D) or parallel (E) configuration, fixed and stained with phalloidin, ~3 hr after excision. Insets: bright-field (top) and zoomed view of the actin organization (bottom). The arrows indicate the alignment of actin fibers that are perpendicular to the wire in the perpendicular configuration and parallel to it in the parallel configuration. See also Figures S1 and S2.

The structural inheritance of the body axis is also reflected in the sensitivity of morphogenesis to the initial tissue geometry. We find that the actin fiber organization at the onset of regeneration, which depends on the initial tissue geometry and the folding process, is correlated with the emerging body plan and the development of morphological defects in the regenerated animals. Fragments and strips typically fold into spheroids, in which the actin fibers have a single preferred orientation. This initial actin organization leads to a normal, single-axis morphology through the canalizing forces generated by the actomyosin cytoskeleton. In contrast, in open and closed rings, the folded spheroids often contain multiple regions with distinct actin fiber alignment, leading to a high incidence of animals with multiple axes and other morphological defects.

The emergence of large-scale coherent actin structures in regenerating *Hydra*, despite the lack of global alignment in the initial spheroid, demonstrates the ability of the actomyosin cytoskeleton to dynamically rearrange during morphogenesis. This observation suggests the presence of mechanical feedbacks as a mechanism for defining and stabilizing a body axis. The significance of mechanical feedbacks is further highlighted by our experiments on regenerating tissues anchored on wires. The mechanical constraints imposed by the wire restrict the dynamic modes of shape variation in anchored tissue, which appear to feed back on the patterning process and significantly increase the tendency of the system to develop an ordered body plan, compared to non-anchored tissues.

We suggest the following mechanism for positive mechanical feedback. Contractions of actin fibers within cells produce dipole-like forces in the tissue, which collectively generate the observed shape deformations. We suggest that the force experienced by a particular cell in a regenerating tissue, collectively created by the cells surrounding it, promotes actin fiber formation and alignment within that cell. Such alignment is well documented in adherent cells, such as fibroblasts, in which stress fibers form an ordered array following the application of an external force (Sears and Kaunas, 2016; Vogel and Sheetz, 2006). We believe that a similar mechanism is in play here, albeit with endogenous force generation in the tissue replacing the external forces applied in vitro. This positive feedback mechanism reinforces actin fiber organization in individual cells and promotes fiber alignment between neighboring cells. We propose that over time, these dynamics stabilize a coherent supra-cellular arrangement of actin fibers in the tissue along a preferred direction. This mechanical self-organization naturally provides canalizing forces that promote the formation of a single body axis, even in the absence of external constraints.

Our findings also touch on the important issue of stability-flexibility duality, which is one of the hallmarks of living systems (Braun, 2015; Fletcher and Mullins, 2010; Soen et al., 2015). This seemingly contradicting duality underlies the ability of the living system to adapt to changing environments, while simultaneously preserving their integrity and stability, even under fluctuating conditions. The actin rearrangement during folding and

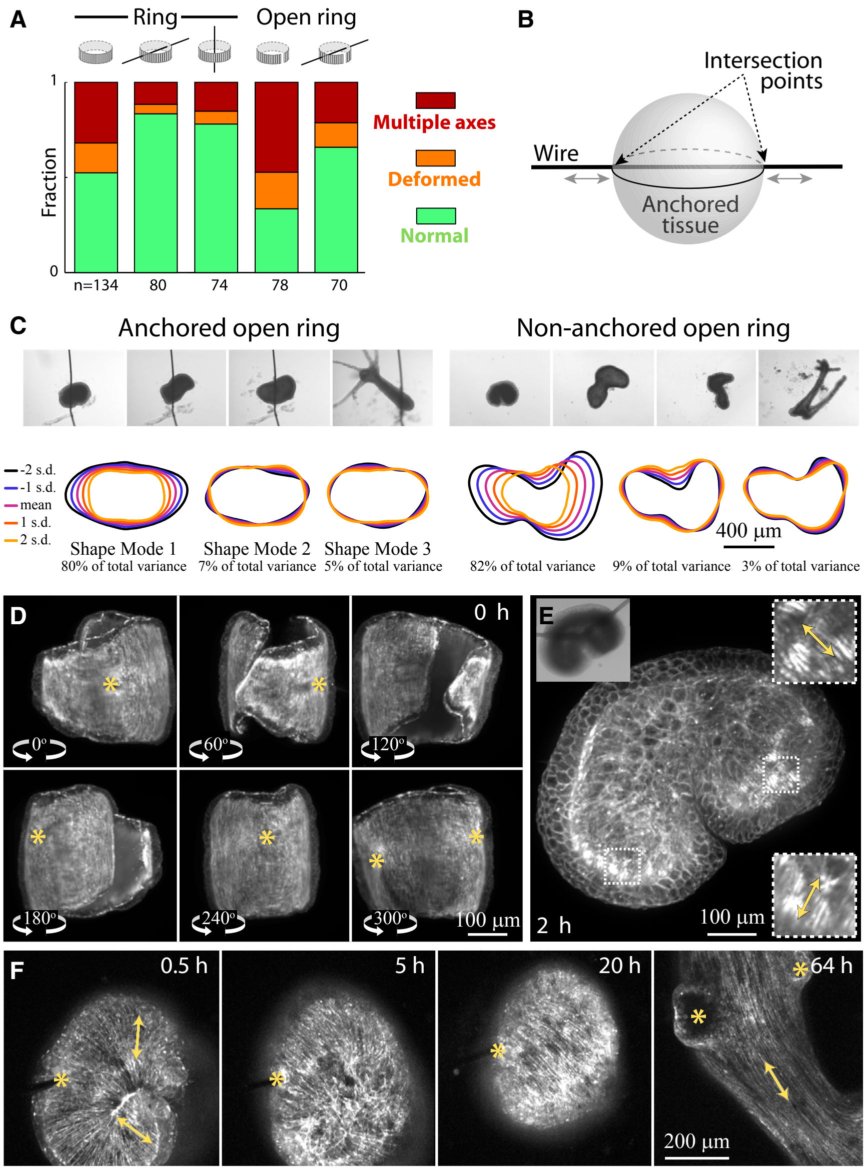


Figure 7. Anchoring Regenerating *Hydra* Tissues on Wires Promotes Order

(A) The influence of an anchoring wire on the regeneration outcome. Bar plot showing the fraction of regenerating *Hydra* in each category of morphological outcome (as in Figure 2F) for closed and open rings in the presence or absence of an anchoring wire. The fraction of rings and open rings that regenerate into animals that are deformed or have multiple body axes is significantly reduced with wires (Chi-squared homogeneity test, p values < 10^{-3}).

(B) Schematic illustration of the mechanical constraints induced by the wire. The stiff wire intersects the hollow tissue spheroid at two points. Tissue movement perpendicular to the wire at the intersection points requires the wire to cut through the tissue layer, so sliding of the tissue along the wire is favored.

(C) Top: bright-field images from time-lapse movies of open rings regenerating in the presence (left) or absence (right) of an anchoring wire (Movie S9). The initial morphology of the folded spheroid appears similar, but the outcome of the regeneration process is different; the non-anchored open ring regenerates into a two-headed animal, whereas the anchored open ring regenerates into an animal with a single body axis. Bottom: the principal modes of shape variation during the first 24 hr of the movies in the presence (right) or absence (left) of an anchoring wire, as determined by principle components analysis of the contours (see Experimental Procedures). The three modes depicted capture >90% of the shape variation over time. For each mode, the mean cell shape is shown together with reconstructions of shapes one and two SDs away from the mean in each direction along the given mode. The variation accounted for by each mode is indicated. The shape modes in the presence of an anchoring wire are considerably more organized.

(D) Maximum projection images of light sheet z stacks of an anchored open ring taken from different angles. The tissue was fixed and stained with AlexaFluor647-phalloidin a few minutes after excision. The yellow asterisks denote the position of the intersection points with the wire.

(E) Maximal projection image of a spinning disk confocal z stack and a bright-field image (upper left) of an open ring anchored on a wire, fixed, and stained with phalloidin Alexa Flour 564 ~2 hr after excision. Zoomed views of the actin organization (right) highlight the different actin fiber alignment in the upper and lower parts of the spheroid, corresponding to the two flaps of the open ring that move in opposite directions as the open ring folds into a spheroid on the wire. The arrows indicate the alignment of actin fibers.

(F) The regeneration of an anchored open ring excised from a transgenic *Hydra* expressing GFP-lifeact in the ectoderm is followed by time-lapse microscopy (Movie S10). Maximum projection images from spinning disk confocal z stacks are shown at different time points. The yellow asterisks denote the position of the intersection points with the wire, and the arrows indicate the alignment of actin fibers. See also Movies S9 and S10.

subsequent regeneration requires high flexibility of the actomyosin system. On the other hand, robust development requires the actomyosin fibers to maintain their integrity and collective ordered state across the entire body throughout the life of an adult *Hydra*. These behaviors can coexist because of the combination of rapid cytoskeletal dynamics and feedback mechanisms; aligned fibers confer a stable memory of the original body axis, whereas the mechanical interactions between disorganized filaments facilitate their dynamic self-organization into a more ordered arrangement of fibers.

Previous work on axis formation in *Hydra* morphogenesis focused primarily on the role of signaling pathways, such as the Wnt system, during head regeneration, budding, and regeneration from aggregates (Bode, 2012; Hobmayer et al., 2000; Lengfeld et al., 2009). Our work shows that the actomyosin cytoskeleton is a significant active player in axis determination in regenerating tissue segments. However, our results do not rule out the participation of other processes in axis determination. Moreover, it is obvious that the actomyosin organization alone is not sufficient to drive differentiation of cells and regeneration of a complete *Hydra*. Thus, we believe that the cytoskeletal dynamics must integrate with other processes, such as the various signaling pathways studied (Bode, 2012; Hobmayer et al., 2000; Lengfeld et al., 2009), to cooperatively determine the body plan of the regenerated animal. How this integration is realized is currently unknown. It is important to note in this context that the non-trivial folding of excised tissue into closed spheroids implies that the body axis in regenerating tissue segments cannot be simply inherited through preservation of the distribution of signaling molecules in the parent animal; any preexisting morphogen gradients in the excised tissue will be deformed by the folding process, which brings together cells originating from different and even opposing regions of the tissue. Thus, the initial spheroid does not have a clear polarity with respect to the body axis of the parent animal.

Our results highlight many important open questions for future work. How does the cytoskeletal organization relate to the spatio-temporal distributions of signaling molecules and cell differentiation? What is the nature of feedbacks arising between the mechanical processes themselves and between them and the morphogenic fields? *Hydra* provides an ideal model organism to investigate these questions and reveals how mechanics and bio-signaling evolve simultaneously as coupled fields. We believe that the lessons learned from this relatively simple system will promote the development of a unified view of morphogenesis that incorporates mechanics as an integral part of the process. Our study, highlighting the role of the actomyosin fiber organization in inheritance and stabilization of the body axis is an initial step in this direction.

EXPERIMENTAL PROCEDURES

Hydra Strains, Culture, and Sample Preparation

Experiments are done with *Magnipapillata Hydra* (kindly supplied by Daniel Sher, Haifa University) or with a transgenic strain of *Hydra Vulgaris* (AEP) expressing lifeact-GFP in the ectoderm (Seybold et al., 2016) for live imaging of actin dynamics (generously provided by Bert Hobmayer, University of Innsbruck). Animals are cultivated in *Hydra* culture medium (1 mM $NaHCO_3$, 1 mM $CaCl_2$, 0.1 mM $MgCl_2$, 0.1 mM KCl, and 1 mM Tris-HCl, pH 7.7) at 18°C. The animals are fed four times a week with live *Artemia* nauplii and washed after ~4 hr. Experiments are initiated ~8–24 hr after feeding. Biochemical perturbations of myosin II activity are done by adding blebbistatin, ML7, Y-27632, or Calyculin A (all from Sigma, dissolved in DMSO) to *Hydra* culture medium. Control experiments are performed in Hydra medium with 0.1% DMSO. The medium is replaced every 24 hr in these experiments to maintain the activity of the inhibitors.

Tissue segments are excised from the middle of a mature *Hydra* using a scalpel equipped with a #15 blade. Rings are excised by performing two nearby transverse cuts. Open rings are generated from rings by making an additional longitudinal cut at one point along the rings' circumference. To obtain fragments, a ring is cut into approximately four parts by additional longitudinal cuts. Longitudinal strips are generated by first cutting a thick ring and then excising thin strips by making longitudinal cuts. Regeneration is defined as the appearance of tentacles, and the regeneration time is defined as the time interval between excision and the appearance of the first tentacle. The size of the tissue segments is parametrized by the effective radius, which is defined as the radius of a circle with the projected area measured after the first shrinkage. For analysis of morphological outcome, we considered tissue pieces with an effective radius in the range of 150–300 μm that regenerated.

Experiments on Wires

We introduce 25-μm-diameter platinum wires (Science Products GmbH) into freshly excised *Hydra* rings. A ring with a wire perpendicular to the original *Hydra* axis is prepared by first skewering the middle of a mature *Hydra* on a wire. The *Hydra* contracts immediately after being pierced, so we wait until the *Hydra* relaxes back to its full length. A ring is subsequently cut by making two transverse incisions on both sides of the wire using a scalpel. An anchored open ring is generated by cutting one side of the ring between the intersection points with the wire. An anchored ring in the parallel configuration is prepared by threading a wire through the hole in the middle of a ring freshly excised from a mature *Hydra*. The anchored tissue pieces with the wires are carefully moved into wells filled with *Hydra* culture medium for imaging, as described below. For analysis of the morphological outcome, we considered only anchored tissue pieces with an effective radius in the range of 150–300 μm that remained on the wire and regenerated.

Hydra Fixation and Actin Staining

Hydra are fixed with 4% formaldehyde in *Hydra* culture medium for ~90 min, followed by three washes in *Hydra* culture medium. The samples are stained using AlexaFluor 546-phalloidin or AlexaFluor 647-phalloidin (Molecular Probes) in 0.5× PBS-BT (0.5× PBS, 1.5% BSA, and 0.05% Triton X-100) for ~60 min. The samples are subsequently washed two times with 0.5× PBS-BT and two times with 0.5× PBS and imaged in 0.5× PBS.

Microscopy

Regenerating tissue fragments are imaged in 96-well plates (for bright-field imaging) or in a 35-mm glass-bottom petri dish (Fluorodish) within small wells made from 2% agarose gel (Sigma) in standard *Hydra* culture medium (for fluorescence imaging). The wells are used to keep the regenerating *Hydra* in place during time-lapse imaging. The agarose samples contain ~20 wells, 3 mm in diameter, in a gel that is 2 mm thick. The sample is further layered with ~5 mL of *Hydra* culture medium.

Time-lapse bright-field movies of regenerating *Hydra* are taken on a Zeiss Axio-observer microscope with a 5× air objective (numerical aperture [NA] = 0.25) at room temperature, and acquired on a charge-coupled device (CCD) camera (CoolSnap, Photometrix).

Spinning disk confocal z stacks are acquired on a spinning disk confocal microscope (Intelligent Imaging Innovations) running Slidebook software using laser illumination and appropriate emission filters at room temperature, and acquired with an EM-CCD (QuantEM; Photometrix). Long time-lapse movies of regenerating *Hydra* expressing lifeact-GFP are taken using a 10× air objective (NA = 0.25). Short time-lapse movies of folding tissue segments expressing lifeact-GFP are taken using a 10× water objective (NA = 0.45). Imaging of fixed *Hydra* are taken using a 10× water objective (NA = 0.45) with a 1.6× optovar.

Fluorescence light sheet microscopy is done on a Lightsheet Z.1 microscope (Zeiss). The light sheet is generated by a pair of 10× air objectives (NA = 0.2) imaged through 20× water objectives (NA = 1) and acquired using a pair of complementary metal-oxide-semiconductor (CMOS) cameras. Fixed samples are embedded in 0.5% agarose gel for imaging. Samples in gel are loaded (prior to solidification) into either a glass capillary with a 1-mm diameter, whereby ~1 cm of the gel tube is extracted into an aqueous medium for imaging, or mounted in a similar diameter fluorinated ethylene propylene (FEP) tubing and imaged through the tubing. Multiple views from different angles are acquired for the same sample by rotating the specimen.

Shape Analysis

Shape analysis of regenerating *Hydra* is done by representing the projected shape of the tissue by polygonal outlines using the Celltool package developed by Zach Pincus (Pincus and Theriot, 2007) and custom code written in MATLAB. Masks depicting the projected tissue shape are determined for a time-lapse movie using the segmentation algorithm described by Seroussi et al. (2012). The shape contours are extracted from the masks to derive a series of (x,y) points corresponding to the tissue boundary. Each series is resampled to 200 points, which were evenly spaced along the boundary. The shape modes of the contours from a time-lapse movie are calculated by principal-component analysis (PCA), as described by Pincus and Theriot (2007).

Statistical Analysis

Pairwise comparison of the distribution of regeneration times in different samples (e.g., different initial tissue geometries or in the presence/absence of wires) are done using a two-sample t test. The reported p values reflect the likeliness of the null hypothesis that the two samples are drawn from the same distribution.

Pairwise comparison of the frequency of the various outcome morphologies in different samples is done using a Chi-square test of homogeneity. The reported p value reflects the likeliness of the null hypothesis that the frequency counts for the different morphological outcomes are distributed equally in the two samples.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Supplemental Experimental Procedures, Figures S1–S4, and Movies S1–S10 and can be found with this article online at http://dx.doi.org/10.1016/j.celrep.2017.01.036.

AUTHOR CONTRIBUTIONS

A.L., L.S.-Z., Y.M.-S., E.B., and K.K. performed the experiments, analyzed the data, and discussed the results. E.B. and K.K. wrote the paper.

ACKNOWLEDGMENTS

We thank Gidi Ben Yoseph for superb technical assistance. We thank Dikla Raz Ben-Aroush for help with imaging. We thank Nitsan Dahan from the LS&E Imaging and Microscopy Unit for help with light sheet microscopy. We thank Bert Hobmayer for generously providing transgenic *Hydra* expressing lifeact and Daniel Sher for the *Hydra* strains. We thank Efi Efrati, L. Mahadavan, Sam Safran, Albrecht Ott, and Bert Hobmayer for discussions. We thank Shimon Marom, Yoav Soen, Enas Abu-Shah, Omri Wurtzel, Natalie Dye, and Naama Brenner for comments on the manuscript.

Received: September 20, 2016
Revised: December 11, 2016
Accepted: January 13, 2017
Published: February 7, 2017

REFERENCES

Beloussov, L.V. (2008). Mechanically based generative laws of morphogenesis. Phys. Biol. *5*, 015009.

Bode, H.R. (2009). Axial patterning in hydra. Cold Spring Harb. Perspect. Biol. *1*, a000463.

Bode, H.R. (2012). The head organizer in Hydra. Int. J. Dev. Biol. *56*, 473–478.

Bode, P.M., and Bode, H.R. (1980). Formation of pattern in regenerating tissue pieces of hydra attenuata: I. Head-body proportion regulation. Dev. Biol. *78*, 484–496.

Bode, P.M., and Bode, H.R. (1984). Formation of pattern in regenerating tissue pieces of Hydra attenuata. III. The shaping of the body column. Dev. Biol. *106*, 315–325.

Bode, H., Berking, S., David, C.N., Gierer, A., Schaller, H., and Trenkner, E. (1973). Quantitative analysis of cell types during growth and morphogenesis in Hydra. Wilhelm Roux'Archiv für Entwicklungsmechanik der Organismen *171*, 269–285.

Braun, E. (2015). The unforeseen challenge: from genotype-to-phenotype in cell populations. Rep. Prog. Phys. *78*, 036602.

Brouzés, E., and Farge, E. (2004). Interplay of mechanical deformation and patterned gene expression in developing embryos. Curr. Opin. Genet. Dev. *14*, 367–374.

Browne, E.N. (1909). The production of new hydranths in hydra by the insertion of small grafts. J. Exp. Zool. *7*, 1–23.

Cummings, S.G., and Bode, H.R. (1984). Head regeneration and polarity reversal inHydra attenuata can occur in the absence of DNA synthesis. Wilhelm Roux's Archives of Developmental Biology *194*, 79–86.

Davidson, L.A. (2012). Epithelial machines that shape the embryo. Trends Cell Biol. *22*, 82–87.

Desprat, N., Supatto, W., Pouille, P.-A., Beaurepaire, E., and Farge, E. (2008). Tissue deformation modulates twist expression to determine anterior midgut differentiation in Drosophila embryos. Dev. Cell *15*, 470–477.

Fletcher, D.A., and Mullins, R.D. (2010). Cell mechanics and the cytoskeleton. Nature *463*, 485–492.

Forgacs, G., and Newman, S.A. (2005). Biological Physics of the Developing Embryo (Cambridge University Press).

Fütterer, C., Colombo, C., Jülicher, F., and Ott, A. (2003). Morphogenetic oscillations during symmetry breaking of regenerating Hydra vulgaris cells. Europhys. Lett. *64*, 137.

Galliot, B. (2012). Hydra, a fruitful model system for 270 years. Int. J. Dev. Biol. *56*, 411–423.

Galliot, B. (2013). Regeneration in Hydra. eLS.

Gierer, A. (2012). The Hydra model - a model for what? Int. J. Dev. Biol. *56*, 437–445.

Gierer, A., and Meinhardt, H. (1972). A theory of biological pattern formation. Kybernetik *12*, 30–39.

Gierer, A., Berking, S., Bode, H., David, C.N., Flick, K., Hansmann, G., Schaller, H., and Trenkner, E. (1972). Regeneration of hydra from reaggregated cells. Nat. New Biol. *239*, 98–101.

Gilbert, S.F. (2000). Developmental Biology (Sinauer Associates, Inc).

Goehring, N.W., and Grill, S.W. (2013). Cell polarity: mechanochemical patterning. Trends Cell Biol. *23*, 72–80.

Guillot, C., and Lecuit, T. (2013). Mechanics of epithelial tissue homeostasis and morphogenesis. Science *340*, 1185–1189.

Heisenberg, C.-P., and Bellaïche, Y. (2013). Forces in tissue morphogenesis and patterning. Cell *153*, 948–962.

Hobmayer, B., Rentzsch, F., Kuhn, K., Happel, C.M., von Laue, C.C., Snyder, P., Rothbächer, U., and Holstein, T.W. (2000). WNT signalling molecules act in axis formation in the diploblastic metazoan Hydra. Nature *407*, 186–189.

Howard, J., Grill, S.W., and Bois, J.S. (2011). Turing's next steps: the mechanochemical basis of morphogenesis. Nat. Rev. Mol. Cell Biol. *12*, 392–398.

Ishihara, H., Martin, B.L., Brautigan, D.L., Karaki, H., Ozaki, H., Kato, Y., Fusetani, N., Watabe, S., Hashimoto, K., Uemura, D., et al. (1989). Calyculin A and okadaic acid: inhibitors of protein phosphatase activity. Biochem. Biophys. Res. Commun. *159*, 871–877.

Kass-Simon, G. (1976). Coordination of juxtaposed muscle layers as seen in Hydra. In Coelenterate Ecology and Behavior, G.O. Mackie, ed. (Springer), pp. 705–714.

Krahe, M., Wenzel, I., Lin, K.-N., Fischer, J., Goldmann, J., Kästner, M., and Fütterer, C. (2013). Fluctuations and differential contraction during regeneration of Hydra vulgaris tissue toroids. New J. Phys. *15*, 035004.

Kücken, M., Soriano, J., Pullarkat, P.A., Ott, A., and Nicola, E.M. (2008). An osmoregulatory basis for shape oscillations in regenerating hydra. Biophys. J. *95*, 978–985.

Lengfeld, T., Watanabe, H., Simakov, O., Lindgens, D., Gee, L., Law, L., Schmidt, H.A., Ozbek, S., Bode, H., and Holstein, T.W. (2009). Multiple Wnts are involved in Hydra organizer formation and regeneration. Dev. Biol. *330*, 186–199.

Lenhoff, S.G., Lenhoff, H.M., and Trembley, A. (1986). Hydra and the birth of experimental biology, 1744: Abraham Trembley's Memoires concerning the Polyps (Boxwood Press).

Meinhardt, H. (1993). A model for pattern formation of hypostome, tentacles, and foot in hydra: how to form structures close to each other, how to form them at a distance. Dev. Biol. *157*, 321–333.

Miller, C.J., and Davidson, L.A. (2013). The interplay between cell signalling and mechanics in developmental processes. Nat. Rev. Genet. *14*, 733–744.

Mueller, J.F. (1950). Some observations on the structure of hydra, with particular reference to the muscular system. Trans. Am. Microsc. Soc. *69*, 133–147.

Munjal, A., and Lecuit, T. (2014). Actomyosin networks and tissue morphogenesis. Development *141*, 1789–1793.

Munjal, A., Philippe, J.-M., Munro, E., and Lecuit, T. (2015). A self-organized biomechanical network drives shape changes during tissue morphogenesis. Nature *524*, 351–355.

Park, H.D., Ortmeyer, A.B., and Blankenbaker, D.P. (1970). Cell division during regeneration in Hydra. Nature *227*, 617–619.

Pincus, Z., and Theriot, J.A. (2007). Comparison of quantitative methods for cell-shape analysis. J. Microsc. *227*, 140–156.

Pouille, P.-A., Ahmadi, P., Brunet, A.C., and Farge, E. (2009). Mechanical signals trigger Myosin II redistribution and mesoderm invagination in Drosophila embryos. Sci. Signal. *2*, ra16.

Saias, L., Swoger, J., D'Angelo, A., Hayes, P., Colombelli, J., Sharpe, J., Salbreux, G., and Solon, J. (2015). Decrease in cell volume generates contractile forces driving dorsal closure. Dev. Cell *33*, 611–621.

Saitoh, M., Ishikawa, T., Matsushima, S., Naka, M., and Hidaka, H. (1987). Selective inhibition of catalytic activity of smooth muscle myosin light chain kinase. J. Biol. Chem. *262*, 7796–7801.

Sarras, M.P., Jr. (2012). Components, structure, biogenesis and function of the Hydra extracellular matrix in regeneration, pattern formation and cell differentiation. Int. J. Dev. Biol. *56*, 567–576.

Sears, C., and Kaunas, R. (2016). The many ways adherent cells respond to applied stretch. J. Biomech. *49*, 1347–1354.

Seroussi, I., Veikherman, D., Ofer, N., Yehudai-Resheff, S., and Keren, K. (2012). Segmentation and tracking of live cells in phase-contrast images using directional gradient vector flow for snakes. J. Microsc. *247*, 137–146.

Seybold, A., Salvenmoser, W., and Hobmayer, B. (2016). Sequential development of apical-basal and planar polarities in aggregating epitheliomuscular cells of Hydra. Dev. Biol. *412*, 148–159.

Shimizu, H., Sawada, Y., and Sugiyama, T. (1993). Minimum tissue size required for hydra regeneration. Dev. Biol. *155*, 287–296.

Soen, Y., Knafo, M., and Elgart, M. (2015). A principle of organization which facilitates broad Lamarckian-like adaptations by improvisation. Biol. Direct *10*, 68.

Soriano, J., Rüdiger, S., Pullarkat, P., and Ott, A. (2009). Mechanogenetic coupling of Hydra symmetry breaking and driven Turing instability model. Biophys. J. *96*, 1649–1660.

Straight, A.F., Cheung, A., Limouze, J., Chen, I., Westwood, N.J., Sellers, J.R., and Mitchison, T.J. (2003). Dissecting temporal and spatial control of cytokinesis with a myosin II Inhibitor. Science *299*, 1743–1747.

Thompson, D.W. (1942). On Growth and Form, Second Edition (Cambridge University Press).

Turing, A.M. (1952). The chemical basis of morphogenesis. Philos. Trans. R. Soc. Lond. B Biol. Sci. *237*, 37–72.

Urdy, S. (2012). On the evolution of morphogenetic models: mechano-chemical interactions and an integrated view of cell differentiation, growth, pattern formation and morphogenesis. Biol. Rev. Camb. Philos. Soc. *87*, 786–803.

Vogel, V., and Sheetz, M. (2006). Local force and geometry sensing regulate cell functions. Nat. Rev. Mol. Cell Biol. *7*, 265–275.

Wanek, N., Marcum, B.A., Lee, H.-T., Chow, M., and Campbell, R.D. (1980). Effect of hydrostatic pressure on morphogenesis in nerve-free hydra. J. Exp. Zool. *211*, 275–280.

West, D.L. (1978). The epitheliomuscular cell of hydra: its fine structure, three-dimensional architecture and relation to morphogenesis. Tissue Cell *10*, 629–646.

Wolpert, L., Clarke, M.R.B., and Hornbruch, A. (1972). Positional signalling along hydra. Nat. New Biol. *239*, 101–105.